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DL - Project no. 2

Speech classification using neural networks

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May 5, 2026



This project focuses on speech classification with the use of neural networks. The main objective of the project is to experiment with different neural network architectures and their hyperparameter and evaluate their performance.

The Speech Classification dataset, which was used for training and testing, includes one-second long recordings of 30 short words, by thousands of different people.

Experimentation outline



In this project we investigate the influence of training hyperparameters and different modeling approaches on the performance of 3 neural networks.

The networks investigated in our project are:

1. Custom Convolutional Neural Network (CNN), containing 3 convolutional blocks,
2. Transformer trained from scratch,
3. pretrained Transformer.

Experimentation process



Stage	Subject	Details
1	Baseline training	Training the 3 default models on the reduced dataset and evaluation to later use as the baseline.
2	Hyperparameter testing	Investigation of influence of different hyperparameters on the model performance. • CNN - n_filters: [16, 32, 64, 128, 256, 512], dropout: [0.0, 0.1, 0.2, 0.3, 0.5], • Transformer - n_layers: [2, 4, 8], n_heads: [4, 8, 16], dropout: [0.0, 0.1, 0.2, 0.3, 0.5], • Pretrained Transformer - strategy: "freeze", "partial", "none"].
3	Model investigation	Investigation of influence of the hyperparameters on the model performance and choosing the best models.
4	Training the chosen models on full dataset	Training the chosen models on the dataset containing all the classes to check if they perform similarly when handling more complex data.
5	Addressing problems with "unknown" class	Testing strategies for trouble with misclassifying "unknown" class.



After completing each stage, the models were evaluated. The features taken into account during model selection included the monotonicity of metric values across epochs, best and final metric values (accuracy, F1-score) and confusion matrices.



All computations were performed using GPU T4 x2 available in Kaggle.

All models were trained using the same :

- loss function (Multiclass Cross Entropy),
- optimizer (ADAM),
- batch size (32),
- number of epochs (25),
- learning rate (0.0005),
- scheduler (ReduceLROnPlateau)

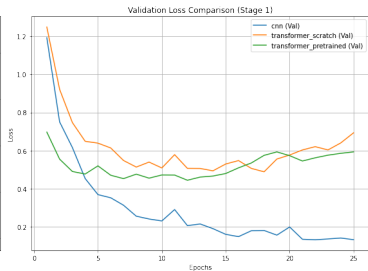
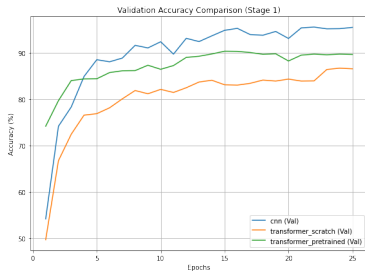
to ensure a sensible comparison environment. To ensure experiment reproducibility, a random seed was fixed.

Results

Stage 1: Baseline models



Model	Best Val F1	Best Epoch	Final Val F1
CNN (baseline)	0.9562	22	0.9553
Transformer (scratch)	0.8677	24	0.8665
Transformer (pretrained)	0.9040	15	0.8969

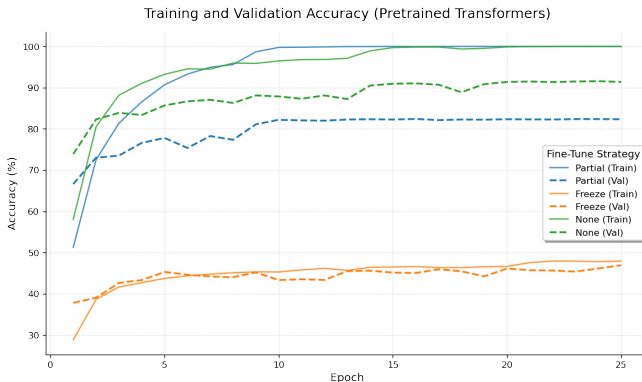


The CNN achieved the highest validation F1 of 0.956, higher than both the scratch-trained transformer (0.868) and the pretrained transformer (0.904). For both transformer models the validation loss curves show clear sign of overfitting. The CNN model, by contrast, maintains a decreasing loss function curve all the way to 25th epoch.

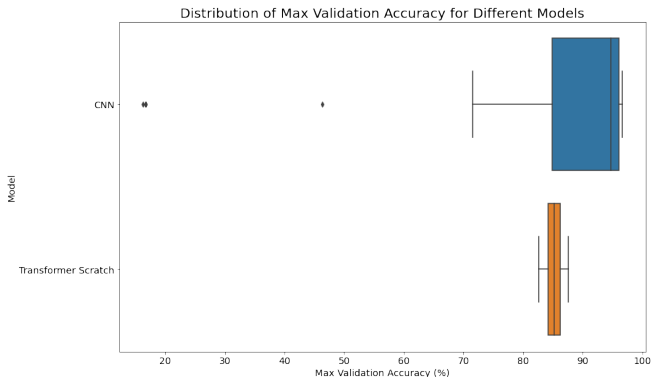
Stage 2: Pretrained transformer



Strategy	Best Val F1	Best Epoch
None (full fine-tuning)	0.9156	24
Partial	0.8242	16
Freeze	0.4627	25



Stage 2: CNN and transformer performance distribution

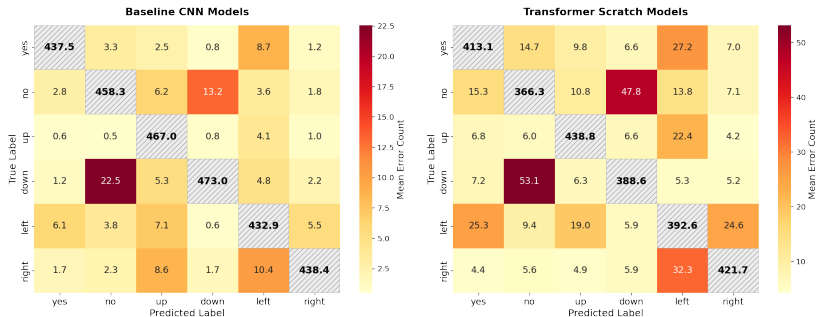


The saturation of hyperparameters may suggest that the architecture has reached a task-specific limit under the given training budget. Likely 25 epochs are insufficient to exploit deeper attention hierarchies at this dataset size. The patterns are better captured by convolutional inductive biases than by global self-attention.

Stage 2: CNN and transformer class distribution



Averaged Confusion Matrices (Models with Best Val F1 > 0.8)



Both architectures had the most problems with classes *down* and *no*.

Stage 3: choice of models for further experiments



CNN:

- CNN-1 (primary),
- CNN-2 (large, no regularisation),
- CNN-3 (small, heavy regularisation).

Transformer (scratch):

- Transformer-1 (primary),
- Transformer-2 (larger network),
- Transformer-3 (primary architecture, no regularization)

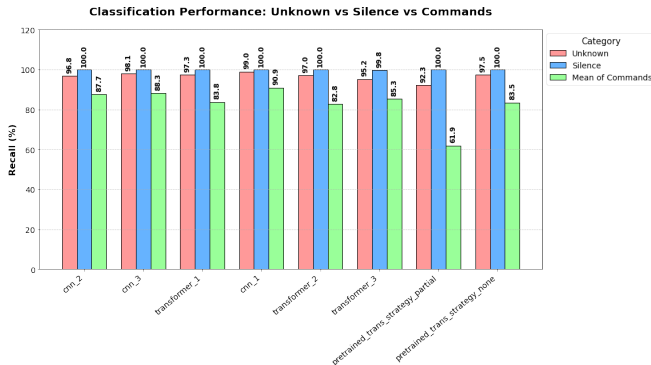
Stage 4: Training chosen models on the entire dataset



Table: Comparison of performance between Stage 2 (6 classes) and Stage 4 (12 classes).

Model	Stage 2	Stage 4 (Full 12-class dataset)				
	Best F1	Best F1	Epoch	Final F1	Train Acc	Gap
CNN-1	0.966	0.9614	19	0.9609	97.72%	+1.54 pp
CNN-2	0.946	0.9365	24	0.9357	98.42%	+4.74 pp
CNN-3	0.954	0.9461	22	0.9332	93.37%	−1.31 pp
Transformer-1	0.876	0.9252	25	0.9252	99.10%	+6.47 pp
Transformer-2	0.873	0.9200	25	0.9200	96.75%	+4.63 pp
Transformer-3	0.872	0.9188	23	0.9163	98.58%	+6.69 pp
Pretrained (none)	0.916	0.9250	22	0.9242	99.88%	+7.26 pp
Pretrained (partial)	0.824	0.8128	24	0.8123	99.97%	+18.09 pp

Stage 5: Issues with the "unknown" and "silence" classes



The reason for this pattern is directly tied to class imbalance. The "unknown" class comprises 61.1% of the validation set. Consequently, when a model is uncertain about a specific command, the observation is statistically much more likely to its features with diverse samples from the unknown class. This was confirmed by the confusion matrix analysis.

Stage 5: Addressing the "unknown" and "silence" classes



Stage	Model / Strategy	Best Val F1	Cmd Recall (mean)
4	CNN-1 (baseline)	0.961	0.909
4	Transformer-1 (baseline)	0.925	0.838
5	CNN-1 + weighted sampling	0.902	0.932
5	Transformer-1 + weighted sampling	0.869	0.842
5	Cascade-1 (CNN)	0.922	0.926
5	Cascade-2 (Transformer)	0.673	0.355

Weighted sampling increases command recall but destroys silence recall, still resulting in a drop in F1. The CNN cascade achieves a similar trade-off to Stage 4 with better command recall, but the F1 score remains below CNN-1 due to persistent silence misclassification. The transformer cascade is not a viable approach, as the transformer cascade filter's tendency to over-predict unknown makes the specialist stage meaningless.



- CNNs are better suited for small spectrogram based tasks due to their built-in locality bias.
- The severe class imbalance in the dataset created a fundamental challenge that was not resolved through used sampling strategy or cascade architectures.
- CNN-1 trained jointly on all 12 classes remains the strongest single model, and the unknown imbalance problem proves to be not solvable with used techniques and project setting constraints.

Thank you for your attention!